

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

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## Device Specifications

Device Type: Matrix Switcher  
Manufacturer: Blackmagic Design  
Firmware Version: N/A  
Model(s): Smart Videohub 12x12, Smart Videohub 20x20, Universal Videohub 288, Smart Videohub 40x40, Universal Videohub 72

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 3.00.0000-b022                         | 2.0.0                   |

## Version History

| Module Version | Date       | Notes                     |
|----------------|------------|---------------------------|
| 1_0_6_1        | 10/17/2018 | Updated module to Rev. B1 |
| 1_0_6_0        | 7/24/2018  | Initial Version           |

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`
- **Call Update method with HeartBeat command to maintain connection status with the device.**

## Supported Classes and Examples

### SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Smart Videohub 12x12')
```

### SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Smart Videohub 12x12')
```

### EthernetClass

```
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9990, Model='Smart Videohub 12x12')
```

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## Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                                          |                                |
|----------------------------------------------------------------------------------------------------------|--------------------------------|
| Command<br><b>MatrixTieCommand</b>                                                                       | Value<br>None                  |
| Qualifier Key<br><b>'Input'</b> <sup>1</sup>                                                             | Qualifier Value<br>'1' – '288' |
| Qualifier Key<br><b>'Output'</b> <sup>2</sup>                                                            | Qualifier Value<br>'1' – '288' |
| # MatrixTieCommand example<br>InterfaceName.Set('MatrixTieCommand', None, {'Input': '1', 'Output': '1'}) |                                |

<sup>1</sup> 'Input' qualifier range is proportional to the number of inputs on the device.

<sup>2</sup> 'Output' qualifier range is proportional to the number of outputs on the device.

## Status Available

For all commands except HeartBeat, Update should be called only once since the command's status will be updated automatically as the device's status changes. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                                                                                                   |                                |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|
| <b>Command</b><br><b>ConnectionStatus</b>                                                                                                                                                                                                                         | Value<br>'Connected'           | Value<br>'Disconnected' |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                                                                                                   |                                |                         |
| <b>Command</b><br><b>HeartBeat</b>                                                                                                                                                                                                                                | Value<br>None                  |                         |
| # HeartBeat examples<br>InterfaceName.Update('HeartBeat')                                                                                                                                                                                                         |                                |                         |
| <b>Command</b><br><b>InputTieStatus</b>                                                                                                                                                                                                                           | Value<br>'Tied'                | Value<br>'Untied'       |
| <b>Qualifier Key</b><br>'Input' <sup>1</sup>                                                                                                                                                                                                                      | Qualifier Value<br>'1' – '288' |                         |
| <b>Qualifier Key</b><br>'Output' <sup>2</sup>                                                                                                                                                                                                                     | Qualifier Value<br>'1' – '288' |                         |
| # InputTieStatus examples<br>InterfaceName.Update('InputTieStatus', {'Input': '1', 'Output': '1'})<br>Value = InterfaceName.ReadStatus('InputTieStatus', {'Input': '1', 'Output': '1'})<br>InterfaceName.SubscribeStatus('InputTieStatus', None, FeedbackHandler) |                                |                         |
| <b>Command</b><br><b>OutputTieStatus</b> <sup>3</sup>                                                                                                                                                                                                             | Value<br>'1' – '288'           |                         |
| <b>Qualifier Key</b><br>'Output' <sup>2</sup>                                                                                                                                                                                                                     | Qualifier Value<br>'1' – '288' |                         |
| # OutputTieStatus examples<br>InterfaceName.Update('OutputTieStatus', {'Output': '1'})<br>Value = InterfaceName.ReadStatus('OutputTieStatus', {'Output': '1'})<br>InterfaceName.SubscribeStatus('OutputTieStatus', None, FeedbackHandler)                         |                                |                         |

- <sup>1</sup> 'Input' qualifier range is proportional to the number of inputs on the device.
- <sup>2</sup> 'Output' qualifier range is proportional to the number of outputs on the device.
- <sup>3</sup> OutputTieStatus range is proportional to the number of inputs on the device.

## Cable and Adapter Requirements

Captive Screw to Male RS-422 DB9 Serial Cable or Captive Screw to Male RJ11 Serial Cable depending on the device

## Notes for the Device

## Serial communication

**Port Type:** RS-422

**Baud Rate:** 9600

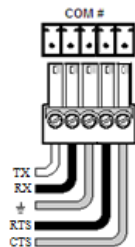
**Data Bits:** 8

**Parity:** None

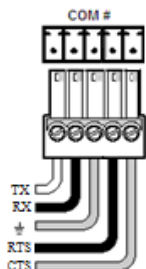
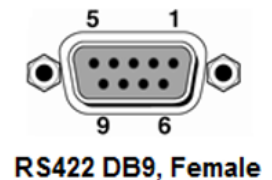
**Stop Bits:** One

**Flow Control:** None

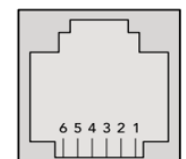
## Pin Assignments Diagram



| Signal | Main Cable | Pin  | Signal |
|--------|------------|------|--------|
| Tx     | →          | 8    | Rx-    |
| Rx     | ←          | 2    | Tx-    |
| GND    | →          | 4, 6 | GND    |
| RTS    | →          | 3    | Rx+    |
| CTS    | ←          | 7    | Tx+    |



| Signal | Main Cable | Pin  | Signal |
|--------|------------|------|--------|
| Tx     | →          | 5    | Rx-    |
| Rx     | ←          | 2    | Tx-    |
| GND    | →          | 3, 4 | GND    |
| RTS    | →          | 6    | Rx+    |
| CTS    | ←          | 1    | Tx+    |



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## Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

|                                       |              |
|---------------------------------------|--------------|
| <b>Port Type:</b>                     | Ethernet     |
| <b>Default Port:</b>                  | 9990         |
| <b>Logon Credentials Supported:</b>   | No           |
| <b>Multi-Connection Capabilities:</b> | Undetermined |
| <b>Port Changeability:</b>            | No           |

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## Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

## Notes for the Device

## Appendix A. Set Commands

|                                                              |                                                    |
|--------------------------------------------------------------|----------------------------------------------------|
| <b>Matrix Tie Command None Input 1 Output 1 (Serial)</b>     | @ X:0/0,0\x0D                                      |
| <b>Matrix Tie Command None Input 1 Output 20 (Serial)</b>    | @ X:0/13,0\x0D                                     |
| <b>Matrix Tie Command None Input 20 Output 1 (Serial)</b>    | @ X:0/0,13\x0D                                     |
| <b>Matrix Tie Command None Input 20 Output 20 (Serial)</b>   | @ X:0/13,13\x0D                                    |
| <b>Matrix Tie Command None Input 1 Output 40 (Serial)</b>    | @ X:0/27,0\x0D                                     |
| <b>Matrix Tie Command None Input 40 Output 1 (Serial)</b>    | @ X:0/0,27\x0D                                     |
| <b>Matrix Tie Command None Input 40 Output 40 (Serial)</b>   | @ X:0/27,27\x0D                                    |
| <b>Matrix Tie Command None Input 1 Output 12 (Serial)</b>    | @ X:0/B,0\x0D                                      |
| <b>Matrix Tie Command None Input 12 Output 1 (Serial)</b>    | @ X:0/0,B\x0D                                      |
| <b>Matrix Tie Command None Input 12 Output 12 (Serial)</b>   | @ X:0/B,B\x0D                                      |
| <b>Matrix Tie Command None Input 1 Output 72 (Serial)</b>    | @ X:0/47,0\x0D                                     |
| <b>Matrix Tie Command None Input 72 Output 1 (Serial)</b>    | @ X:0/0,47\x0D                                     |
| <b>Matrix Tie Command None Input 72 Output 72 (Serial)</b>   | @ X:0/47,47\x0D                                    |
| <b>Matrix Tie Command None Input 1 Output 288 (Serial)</b>   | @ X:0/11F,0\x0D                                    |
| <b>Matrix Tie Command None Input 288 Output 1 (Serial)</b>   | @ X:0/0,11F\x0D                                    |
| <b>Matrix Tie Command None Input 288 Output 288 (Serial)</b> | @ X:0/11F,11F\x0D                                  |
| <b>Matrix Tie Command None Input 1 Output 1 (Ethernet)</b>   | VIDEO OUTPUT ROUTING:\x0D\x0A0 0\x0D\x0A\x0D\x0A   |
| <b>Matrix Tie Command None Input 1 Output 20 (Ethernet)</b>  | VIDEO OUTPUT ROUTING:\x0D\x0A0 19\x0D\x0A\x0D\x0A  |
| <b>Matrix Tie Command None Input 20 Output 20 (Ethernet)</b> | VIDEO OUTPUT ROUTING:\x0D\x0A19 19\x0D\x0A\x0D\x0A |
| <b>Matrix Tie Command None Input 72 Output 72 (Ethernet)</b> | VIDEO OUTPUT ROUTING:\x0D\x0A71 71\x0D\x0A\x0D\x0A |

## Appendix B. Update Commands

|                                                      |                                        |
|------------------------------------------------------|----------------------------------------|
| <b>HeartBeat (Serial)</b>                            | \x0D                                   |
| <b>HeartBeat (Ethernet)</b>                          | PING: \x0D\x0A\x0D\x0A                 |
| <b>Input Tie Status Input 1 Output 1 (Serial)</b>    | @ X?00\x0D                             |
| <b>Input Tie Status Input 1 Output 12 (Serial)</b>   | @ X?0B\x0D                             |
| <b>Input Tie Status Input 1 Output 288 (Serial)</b>  | @ X?011F\x0D                           |
| <b>Output Tie Status Output 1 (Serial)</b>           | @ X?00\x0D                             |
| <b>Output Tie Status Output 12 (Serial)</b>          | @ X?0B\x0D                             |
| <b>Output Tie Status Output 288 (Serial)</b>         | @ X?011F\x0D                           |
| <b>Input Tie Status Input 1 Output 1 (Ethernet)</b>  | VIDEO OUTPUT ROUTING: \x0D\x0A\x0D\x0A |
| <b>Input Tie Status Input 1 Output 12 (Ethernet)</b> | VIDEO OUTPUT ROUTING: \x0D\x0A\x0D\x0A |
| <b>Input Tie Status Input 1 Output 288(Ethernet)</b> | VIDEO OUTPUT ROUTING: \x0D\x0A\x0D\x0A |